Unbalanced MZI for Group Index Extraction

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***Abstract*—The goal of this work is to extract the group index of a quasi-TE mode in a silicon strip waveguide using unbalanced Mach-Zehnder interferometers (MZIs). For this, the MZIs were simulated, designed and experimentally characterized after fabrication. The experimentally obtained values of the group index range from 4.183 to 4.232 at 1550 nm and agree with simulations, with deviations below 0.82%. Devices with the same path length difference show a reproducibility of approximately 0.10%. A comparison between two measurement setups shows relative deviations below 0.2%.**

I. Introduction

Silicon photonics is a key enabling technology for high-bandwidth optical interconnects in data centers and high-performance computing [1]. It enables the integration of multiple functionalities in a single chip making use of the advanced fabrication technologies developed for micro-electronics promising high-complexity at low cost [2]. The Mach–Zehnder interferometer (MZI) specifically is an important building block for optical switches and modulators in silicon photonics [3], [4]. They operate based on converting a phase shift between two optical paths into an intensity change at the output. They can further be utilized to extract waveguide parameters such as the group index n_g and the waveguide dimensions [5]. In this work, multiple imbalanced MZIs with different path length differences ΔL between 30 µm and 190 µm will be designed and characterized to extract the group index of a quasi-TE mode in a silicon strip waveguide with 220 nm height and 500 nm width on silicon oxide. To check the fabrication reproducibility of the proposed photonic circuits, one interferometer design with ΔL= 90 µm will be fabricated three times. The group index will be calculated using the free spectral range (FSR) of the interferometer and be compared to simulations.

First, the theoretical background of the MZI will be discussed. Then, the waveguide will be simulated using Lumerical MODE to create a compact waveguide model. After that, the MZI will be constructed and simulated using Lumerical INTERCONNECT. Then, the proposed layout and fabrication will be discussed. After fabrication, the circuit will be experimentally characterized and the results will be compared to the simulations.

II. Theory

In this section, the transfer function of an unbalanced MZI will be discussed following calculations from [2]. From that, the free spectral range (FSR) of the MZI will be calculated and the formula used for extracting the group index will be discussed.

The unbalanced MZI consists of an input that is split into two branches including two waveguides of different lengths and then recombined. In this work, Y-branches are used to split and later recombine the signal.

At the first Y-branch, the input intensity $I_i \sim |E_i|^2$ is split equally to the upper and lower branch of the interferometer with E_i describing the electric field. Therefore, both branches have the field $E_i/\sqrt{2}$ after the first Y-branch.

The propagation of the light through the waveguides is described by the propagation constant $\beta = \frac{2\pi n_{eff,j}(\lambda)}{\lambda} + i\frac{\alpha_j}{2}$ with wavelength λ, effective index n_j and loss α_j of the respective waveguide with length L_j. Neglecting losses ($\alpha \approx 0$), the electric field at the end of each waveguide $E_{o,j}$ is

$$E_{0,j} = \frac{E_i}{\sqrt{2}} e^{-i\beta_j L_j}, j = 1, 2. \tag{1}$$

The two outputs are then combined at the second Y-branch so that the output at the end of the MZI can be described as

$$E_o = \frac{E_{o,1} + E_{o,2}}{\sqrt{2}} = \frac{E_i}{2}(e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}), \tag{2}$$

or in terms of the intensity

$$\frac{I_o}{I_i} = \frac{1}{4} \cdot |e^{-i\beta_1 L_1} + e^{-i\beta_2 L_2}|^2 = \frac{1}{2}(1 + \cos(\beta_1 L_1 - \beta_2 L_2)). \tag{3}$$

Due to the wavelength-dependence of the propagation constant, the output of the MZI is therefore a wavelength-dependent sinusoidally varying function. The period of this function is the FSR of the interferometer. Using the same waveguide material and dimensions ($\beta_1 = \beta_2$) in an unbalanced MZI ($L_1 \neq L_2$) with path length difference $\Delta L = L_2 - L_1$, equation (3) reduces to

$$\frac{I_o}{I_i} = \frac{1}{2}(1 + \cos(\beta(\lambda) \cdot \Delta L)). \tag{4}$$

The $FSR = \Delta\lambda = \lambda_{m+1} - \lambda_m$ can then be calculated knowing that the phase shift between two adjacent peaks is $\beta_m \Delta L - \beta_{m+1}\Delta L = 2\pi$. Assuming β varies linearly in λ, one can find $\frac{d\beta}{d\lambda} = -\frac{2\pi}{\lambda^2}(n - \frac{dn}{d\lambda}\lambda)$ [6]. The FSR can therefore be calculated using

$$FSR = \frac{\lambda^2}{\Delta L(n - \lambda\frac{dn}{d\lambda})} = \frac{\lambda^2}{\Delta L n_g}. \tag{5}$$

Finally, the group index n_g can be calculated for known path differences ΔL after measuring the FSR of the proposed MZIs according to

$$n_g = n - \lambda\frac{dn}{d\lambda} = \frac{\lambda^2}{FSR \cdot \Delta L}. \tag{6}$$

In the experiment, the group index will be calculated by measuring the spectrum of the MZI, identifying adjacent maxima and computing the FSR. The latter can be averaged

over multiple fringes to reduce the error and then inserted into equation (6).

III. Modeling and Simulation

Lumerical MODE [7] is used to extract the wavelength dependent effective index and group index of the silicon waveguide and develop a compact model for the waveguide. The data will then be exported to Lumerical INTERCONNECT [8] for the simulation of the optical circuit.

A. Waveguide simulations using Lumerical MODE

The waveguide-width was set to 500 nm and the height was set to 220 nm. A mesh convergence study was performed for the FDE mode solver. The effective index changed from 2.503380 (100×100) to 2.501430 (300×300) and 2.501311 (500×500) for the quasi-TE-mode at 1550 nm. Since the change between 300×300 and 500×500 was $1.19 \cdot 10^{-4}$, the 500×500 mesh was used for all reported results. The simulated mode profile of the fundamental quasi-TE-mode is shown in figure 1.

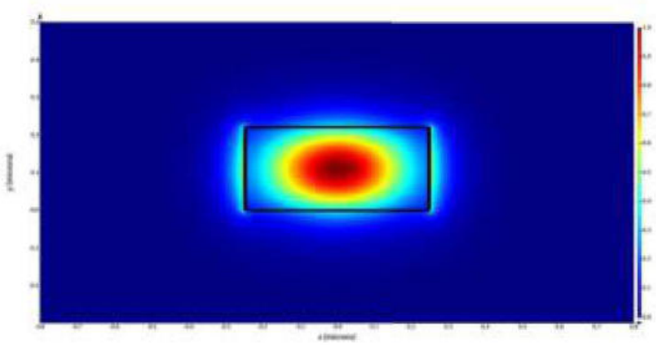

Fig. 1. Simulated waveguide mode profile of the fundamental quasi-TE mode of the 500 nm x 220 nm silicon waveguide at 1500 nm.

Then, a wavelength sweep between 1500 nm and 1600 nm including material and waveguide dispersion was performed. The resulting effective index and group index are shown in figure 2 and figure 3.

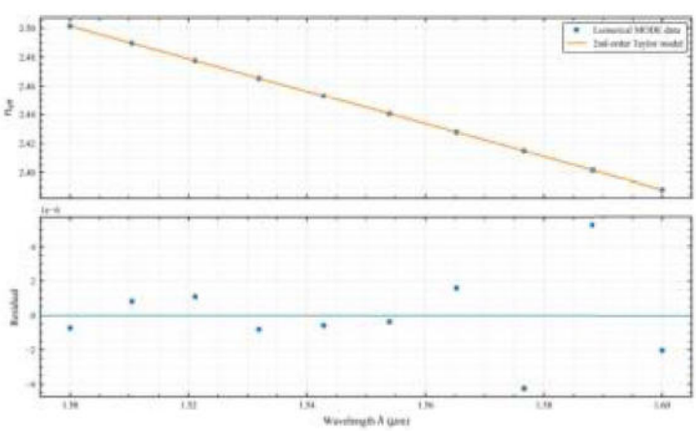

Fig. 2. Effective index n_{eff} calculated in Lumerical MODE and fit following equation (7) vs. wavelength for the fundamental quasi-TE mode (1500–1600 nm).

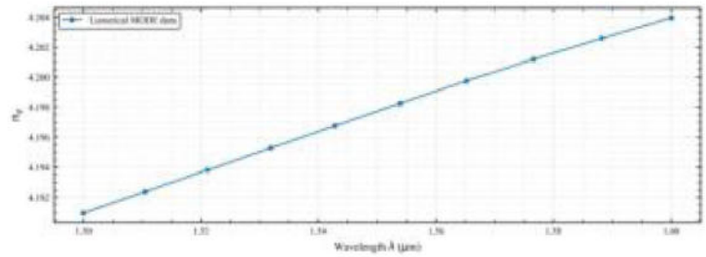

Fig. 3. Group index n_g vs. wavelength for the fundamental quasi-TE mode (1500–1600 nm). The fitted group index at 1550 nm (fit uncertainty only) is 4.197599±0.000004.

Afterwards, the compact model for the waveguide was derived according to [2]

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \tag{7}$$

Fitting the data directly in Python and inserting them into equation (7) for $\lambda_0 = 1550$ nm and for the first quasi-TE-mode leads to

$$n_1 = 2.4448790 \pm 0.0000014, \tag{8}$$
$$n_2 = (-1.13079 \pm 0.00003)\,\mu\text{m}^{-1}, \tag{9}$$
$$n_3 = (-0.0430 \pm 0.0010)\,\mu\text{m}^{-2}, \tag{10}$$
$$\lambda_0 = 1.55\ \mu\text{m}. \tag{11}$$

Uncertainties reported here reflect the statistical uncertainty of the polynomial regression (fit covariance only). They do not include fabrication tolerances or simulation/model uncertainty. For readability, the coefficients are rounded leading to

$$\begin{aligned} n_{\text{eff}}(\lambda) = 2.445 - 1.131\ \mu\text{m}^{-1}(\lambda - \lambda_0) \\ - 0.043\ \mu\text{m}^{-2}(\lambda - \lambda_0)^2, \end{aligned} \tag{12}$$

with λ and λ_0 in μm.

B. MZI simulation using Lumerical INTERCONNECT

The MZI was constructed in Lumerical INTERCONNECT as shown in figure 4. The MZI consists of two Y-branches (ebeam_y_1550) whose S-parameters were taken from SiEPIC EBEAM INTERCONNECT Library and two waveguides which were imported from Lumerical MODE. The length of waveguide one was set to 100 µm whereas the length of waveguide two was varied between 130 µm and 300 µm to analyze the FSR for ΔL between 30 µm and 190 µm as shown in table I and in figure 5. The fit-only uncertainty of the calculated FSR is sub-pm, which is below the wavelength resolution of the characterization laser (10 pm) and therefore not included here. Furthermore, two grating couplers (GC_TE_1550_8degOxide_BB) from the SiEPIC EBEAM INTERCONNECT Library were added. An optical network analyzer (ONA) was included to display the expected FSR and transmission spectra of the device shown in figures 6 and 7.

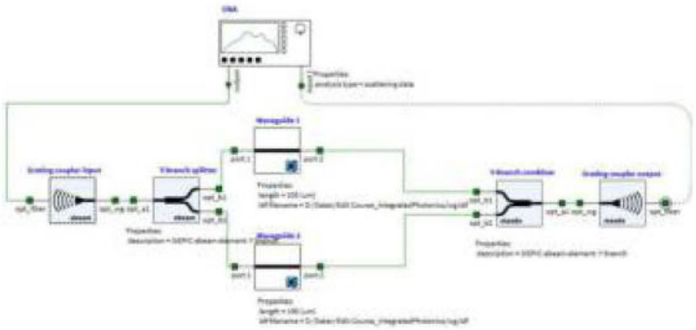

Fig. 4. Exemplary MZI-model implemented in Lumerical INTERCONNECT including the two grating couplers and the optical network analyzer (ONA) used to evaluate the circuit.

TABLE I
EXPECTED FSR AT 1550 NM FOR DIFFERENT ΔL. THE FSR WAS CALCULATED USING EQUATION (5).

ΔL (µm)	FSR (nm)
30	19.05
60	9.52
90	6.35
120	4.76
150	3.81
190	3.01

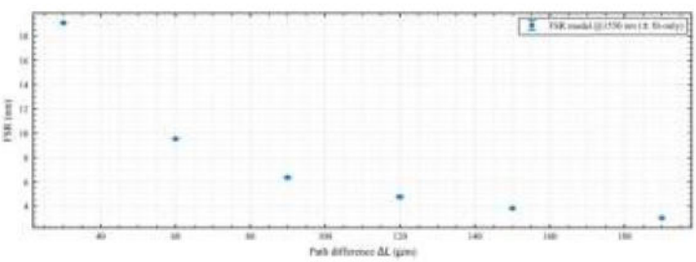

Fig. 5. Calculated FSR vs. wavelength using equation (5) and the path length differences depicted in table I for the fundamental quasi-TE mode. The corresponding uncertainties were obtained by Gaussian error propagation using the fit covariance matrix only. For the final comparison with measured data, uncertainties will be estimated using corner analysis.

Figure 6 shows two exemplary MZI transmission spectra of the quasi-TE mode for 90 µm and 200 µm path difference. As expected, the FSR decreases for larger path length differences.

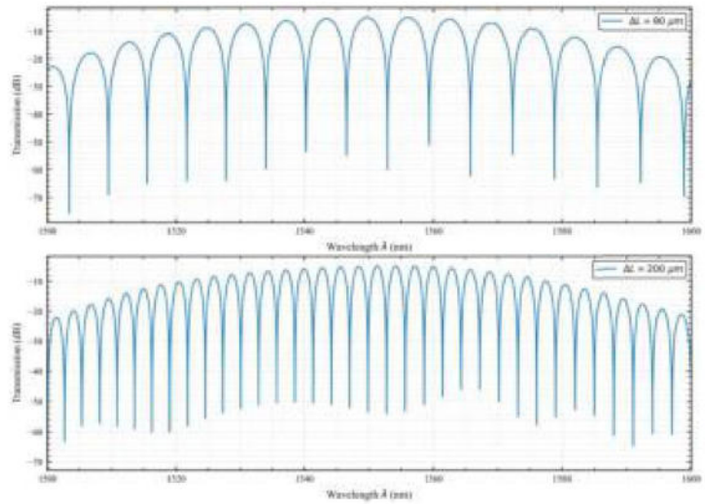

Fig. 6. Exemplary transmission spectra of the proposed MZI with (top) 90 µm and (bottom) 200 µm path difference.

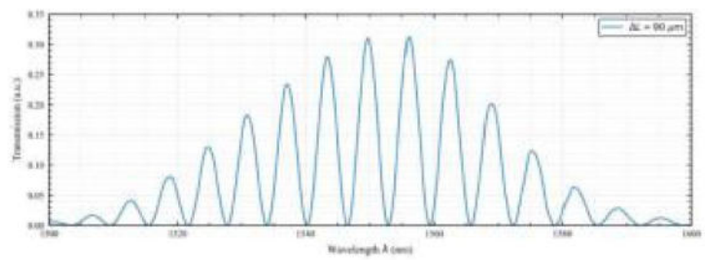

Fig. 7. Exemplary MZI transmission of the quasi-TE mode for ΔL=90 µm in arbitrary units.

TABLE II
DEVICE NAME VS. PATH LENGTH DIFFERENCE ΔL

Device Name	ΔL (μm)
CAL1	N/A, used for calibration
DE	N/A, used for calibration
MZI0	0
MZI1	30
MZI2	60
MZI3	90
MZI4	90
MZI5	90
MZI6	120
MZI7	150
MZI8	190

IV. LAYOUT

The devices proposed in Table I were designed in KLayout using the SiEPIC version 0.5.31 and EBeam - v0.4.53. An overview of the layout is shown in figure 8 and described in table II. Three devices were added compared to table I: CAL1 consists of only one Y-branch and three grating couplers. It is used to evaluate the splitting ratio of the Y-branch. DE consists of two grating couplers connected by a waveguide to characterize the grating couplers for calibration. MZI0 is a MZI with no path length difference, again for calibration.

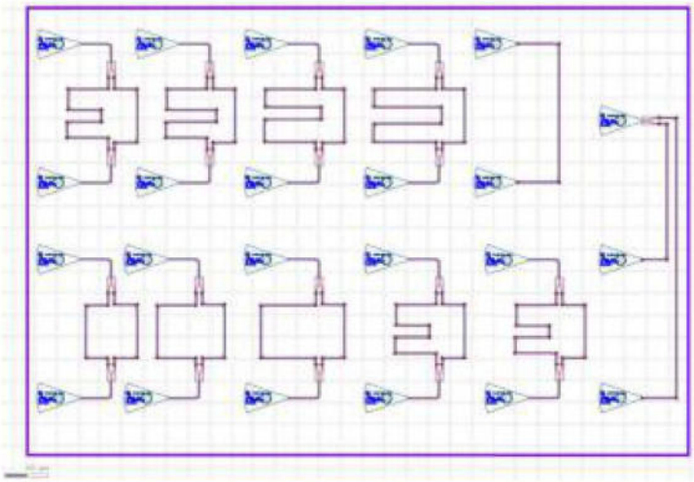

Fig. 8. Proposed layout on the allocated space of 410 µm x 605 µm. Top row from left to right: MZI5, MZI6, MZI7, MZI8, DE. Bottom row from left to right: MZI0, MZI1, MZI3, MZI4 and CAL.

V. Fabrication Process

The devices are fabricated at Applied Nanotools using the NanoSOI MPW fabrication process which is based on direct-write electron beam lithography (EBL) with a 100 keV electron beam [9]. Pre-diced silicon-on-insulator wafers with 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. After an initial wafer clean using piranha solution (3:1 H_2SO_4:H_2O_2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium hydroxide (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. After etching, the remaining device structures are covered with 2.2 µm oxide cladding for protection. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

VI. Fabrication Variability and Corner Analysis

Fabrication uncertainties arise not only from electron beam lithography (EBL), but also from variation of the wafer that is processed. The devices in this work are fabricated on silicon-on-insulator (SOI) wafers provided by Soitec (Grenoble, France) [18]. The wafer specifications include a silicon device layer thickness of 219.2 nm with a 6σ variation of 23.4 nm (corresponding to ± 3.9 nm for 1σ), and a buried oxide thickness of 2994.5 nm with a 6σ variation of 6.3 nm [11]. Furthermore, imperfect sidewall angles occur due to fabrication limitations. These variabilities significantly impact photonic device performance. Silicon thickness deviations on the order of ± 5 nm have been shown to induce shifts in the central wavelength of waveguide Bragg gratings [16]. Similar effects can be expected for unbalanced MZIs, where variations in effective and group index lead to spectral shifts and changes in the free spectral range [11].

To systematically assess the impact of fabrication variability on the circuit performance, a corner analysis is conducted. In this approach, the waveguide width and height, are varied according to the minimum and maximum deviations given by the manufacturer. Furthermore, a nominal design is simulated for reference. The circuit response is then evaluated for all combinations of these parameter sets, representing process corners. This allows for the identification of worst-case performance and provides bounds on expected device behavior under fabrication tolerances [11]. For the devices discussed here, the height is varied between 215.3 nm and 223.1 nm with a nominal value of 220 nm and the widths of the waveguides are simulated between 470 nm and 510 nm with a nominal value of 500 nm. To evaluate the worst-case disparities between the ideal and fabricated design, four corners are analysed: 470 nm x 223.1 nm, 470 nm x 215.3 nm, 510 nm x 215.3 nm and 510 nm x 223.1 nm as well as the nominal design. The corners are simulated numerically in Lumerical MODE and a frequency sweep is conducted for the fundamental quasi-TE mode of the waveguide for each configuration to account for wavelength dependence of the effective index and group index. The results of the corner analysis of the group index are shown in figure 9. The simulated group index at 1550 nm is

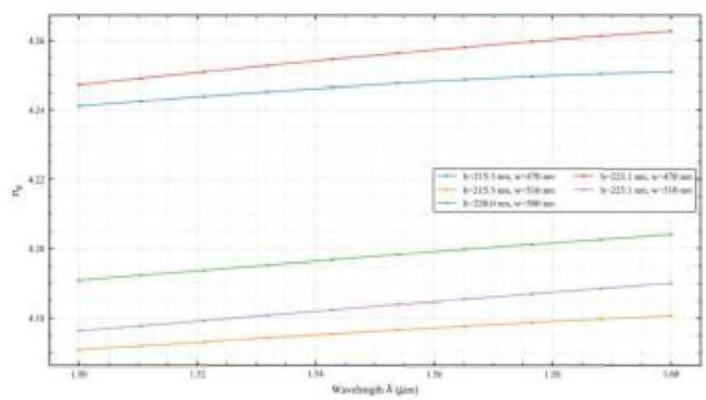

Fig. 9. Corner analysis of the waveguide including all combinations of min/max values given by the manufacturer and the nominal design at 500 nm x 220 nm.

between 4.18 and 4.26. While corner analysis provides worst-case performance bounds, statistical variations can be further evaluated using Monte Carlo method.

VII. Measurement

To characterize the devices, a custom-built automated test setup with automated control software was used [11], [15] [12]. An Agilent 81600B tunable laser was utilized as the

input source and Agilent 81635A optical power sensors as the output detectors. Its wavelength was swept from 1500 nm to 1600 nm in 10 pm steps to characterize the wavelength-dependent response of the devices. A polarization maintaining fibre array and fibres were used to couple in and out of the chip via the grating couplers [13] [14]. Two stages were used for this characterization: The main stage used in this report is the Scylla stage unless labeled otherwise. Furthermore, the devices were measured with a new measurement stage called Iris, developed by Mateo Branion-Calles at the University of British Columbia. The results of both stages will be evaluated and briefly compared later.

VIII. Experimental Data and Analysis

A. Calibration and De-embedding

Two de-embedding structures are used: a reference waveguide (DE) and a balanced Mach–Zehnder interferometer (MZI0). The DE structure replaces the functional device with a single waveguide and serves as a baseline to characterize wavelength-dependent coupling and propagation losses, i.e. due to the grating couplers. MZI0 includes the grating couplers, two Y-branches as well as the same lower interferometer arm geometry as the unbalanced MZIs and is therefore used as the primary calibration structure. Although MZI0 is designed with zero path length difference, residual imbalances may occur due to fabrication variations, leading to deviations from an ideal flat response. Therefore, the DE structure is used as an additional reference to validate the de-embedding approach. Fig. 10 shows a comparison of both de-embedding structures. The MZI0 response (orange) exhibits higher losses compared to the DE structure (blue), as expected due to the additional components. However, the overall spectral shape between 1520 nm and 1580 nm remains largely identical, confirming that both structures capture similar wavelength-dependent effects of the grating couplers. In the following, the MZI0 structure is used for all devices to compensate for losses and to isolate the intrinsic device response. It should however be noted that, due to fabrication variability, the MZI0 structure may exhibit residual phase imbalance.

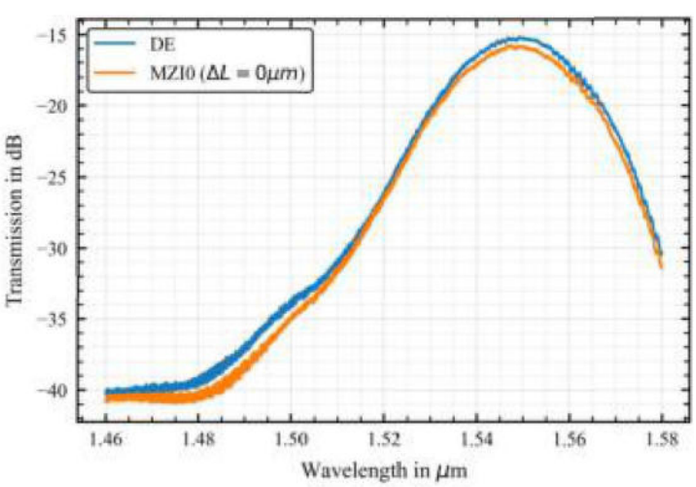

Fig. 10. Comparison of the two de-embedding structures DE (blue) and MZI0 (orange).

To apply the calibration to the other datasets, a low-order polynomial is fitted to the measured calibration spectrum as shown in figure 11. The resulting fit data is subsequently subtracted from the device spectra, resulting in a flattened response. To reduce the influence of measurement noise, the fitting procedure is restricted to the upper portion of the spectrum (approximately the top 10 dB below the maximum), where the signal-to-noise ratio is highest [17]. The inset in figure 11 shows the fit in the selected region. The resulting usable wavelength range for Scylla is 1521.47 nm to 1574.10 nm. For the Iris stage, the wavelength range is 1527.19 nm to 1578.24 nm. Within these wavelength ranges, the Y-branches, characterized using CAL1 exhibit a splitting imbalance from 0.05 dB to 1.65 dB (median 0.24 dB) with 90% of the values below 0.91 dB, indicating a deviation from ideal 50:50 splitting which is also partly due to the different waveguide lengths connecting the Y-Branch to the grating couplers. Imperfect splitting ratio affects the fringe visibility but does not affect the free spectral range in this configuration. Its impact on the extracted group index is therefore negligible compared to other manufacturing variations, although the fit stability may be slightly reduced.

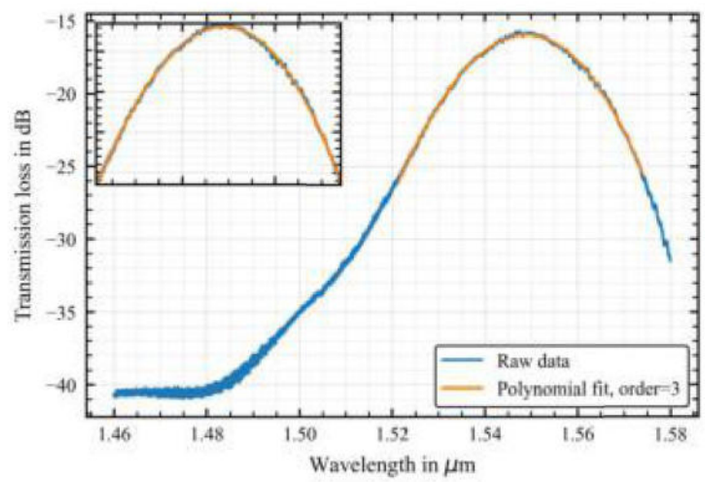

Fig. 11. Polynomial fit (orange) of the transmission of MZI0 (raw data in blue) used for calibration of the following datasets. The order of polynomial is three. The inset shows a zoom-in in the fit data with transmission below -10 dB from the maximum.

B. Data Evaluation

Following calibration, the corrected transmission spectra are analyzed to extract the group index of the waveguides. Fig. 13 shows an example of the calibrated spectrum for MZI7, where the loopback fit obtained from MZI0 has been applied to remove the wavelength-dependent baseline. Subsequently, the spectral periodicity of each MZI is evaluated by detecting adjacent minima in the corrected transmission spectrum. The free spectral range (FSR) is determined from the spacing between neighboring minima and used to calculate the group index according to equation (6). The peak-based estimate is then used as an initial parameter set for fitting the full MZI transfer function to the measured spectrum, as shown in Fig. 15. In addition, autocorrelation is evaluated as a

secondary diagnostic for the spectral periodicity to verify the extracted FSR and n_g. Finally, the experimentally obtained group index is compared to numerical simulation results derived from corner analysis using the Lumerical MODE solver. A frequency sweep of the fundamental quasi-TE mode is performed to extract the wavelength-dependent group index for the considered geometry variations and compared to the experimental data as shown in figure 16. The brown line, indicating the experimentally extracted values lies within the group index range simulated using corner analysis. It lies between the nominal design and the corner with 223.1 nm height and 510 nm width indicating a slightly larger waveguide than in the nominal design. The analysis was done for all MZIs and the results are shown in table III and compared to the simulated group index range from the corner analysis. Furthermore, it was checked which corner was the closest compared to the experimentally extracted values to assess trends across the wafer area. All measured devices lie within the range predicted by the corner analysis, confirming that the observed variations are consistent with fabrication tolerances.

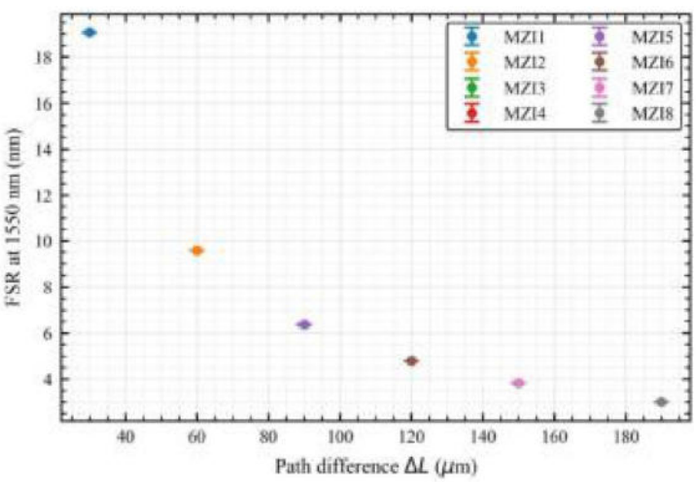

Fig. 12. Experimentally extracted FSRs of MZI1-MZI8.

MZI2, MZI3 and MZI7 exhibit group indices slightly below the nominal value of $n_g = 4.1976$ and indicate slightly wider and taller waveguides than in the nominal case. MZI8 however indicates a smaller waveguide which could either indicate a wafer inhomogeneity or worse calibration as it is situated the furthest away from the calibration structure. The rest of the MZIs are closest to the simulated nominal design. The FSRs of all devices are shown in figure 12. As expected, the FSR decreases for larger path length differences.

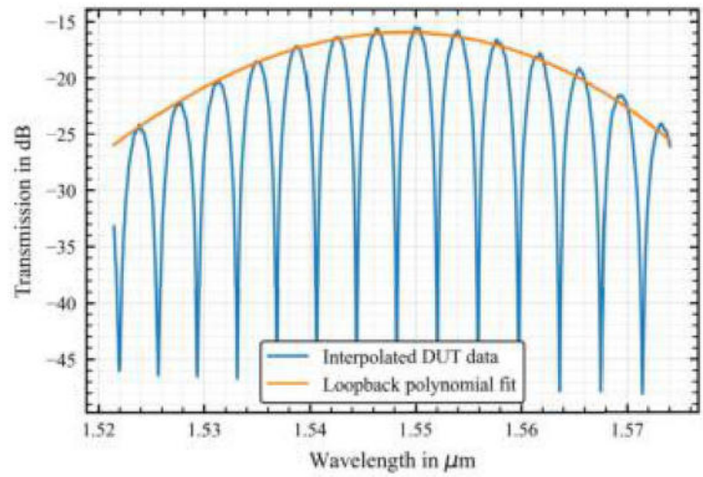

Fig. 13. MZI7 with loopback fit (orange) extracted from MZI0.

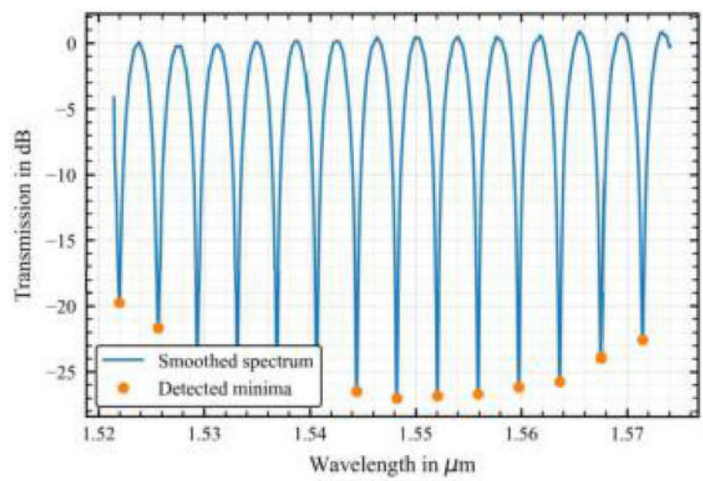

Fig. 14. Peak detection on the corrected MZI7-spectrum from figure 13.

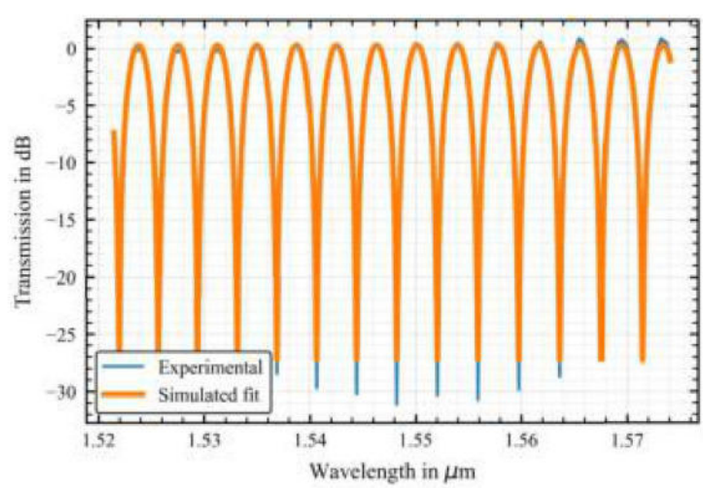

Fig. 15. Fitted MZI transfer function for FSR and n_g extraction vs. experimental data for MZI7.

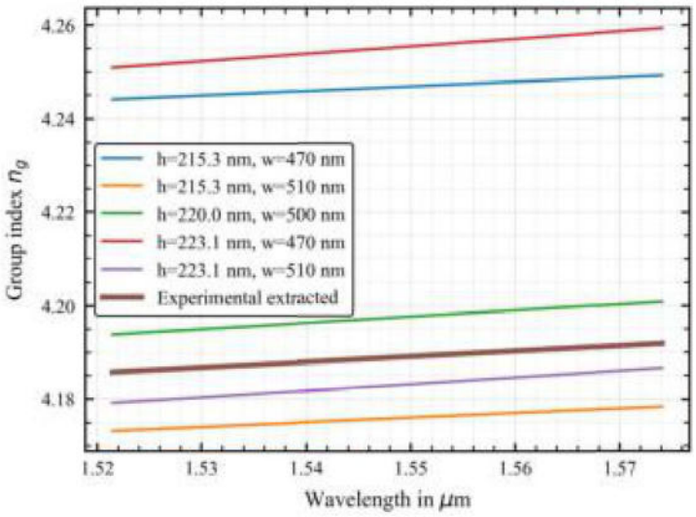

Fig. 16. Extracted n_g from the experimental data (brown) compared to the data gained from numerical simulations doing corner analysis.

TABLE III
COMPARISON OF EXPERIMENTALLY EXTRACTED GROUP INDICES AT 1550 NM (SCYLLA). THE VARIANCE IS OBTAINED FROM THE FIT COVARIANCE. W/IN CI INDICATES WHETHER THE VALUE LIES WITHIN THE CORNER ANALYSIS RANGE.

Device	ΔL (μm)	$n_{g,\text{exp}}$	w/in CI	Nearest corner (h, w)
MZI1	30	4.2008 ± 0.0004	YES	(220.0, 500.0)
MZI2	60	4.1832 ± 0.0003	YES	(223.1, 510.0)
MZI3	90	4.1876 ± 0.0002	YES	(223.1, 510.0)
MZI4	90	4.1915 ± 0.0002	YES	(220.0, 500.0)
MZI5	90	4.1917 ± 0.0002	YES	(220.0, 500.0)
MZI6	120	4.1921 ± 0.0002	YES	(220.0, 500.0)
MZI7	150	4.1891 ± 0.0002	YES	(223.1, 510.0)
MZI8	190	4.2322 ± 0.0002	YES	(215.3, 470.0)

C. Comparison of MZIs with the same path length difference

Three nominally identical interferometers with $\Delta L = 90\,\mu m$ (MZI3–MZI5) were used to evaluate fabrication reproducibility. The extracted group indices are $n_g = 4.1876$, 4.1915, and 4.1917, resulting in a spread of 0.0041 ($\sim 0.10\%$).

The relative deviation between MZI4 and MZI5 is 0.005%, while MZI3 deviates by -0.093% and -0.098% from MZI4 and MZI5, respectively.

In the layout, MZI3 and MZI4 are located directly next to each other, whereas MZI5 is placed in the upper row. Since MZI4 and MZI5 show negligible deviation despite their spatial separation, no systematic position-dependent variation across the chip is observed.

Overall, the small relative spread confirms good reproducibility and demonstrates that the method resolves sub-0.1% device-to-device variations in the tested configuration.

D. Comparison Scylla and Iris stage

The calibration and analysis described in Sections VIII-A and VIII-B were also applied to the Iris dataset and compared with the Scylla results. The extracted group indices are listed in Table IV. The Iris and Scylla datasets show good agreement, with relative deviations below 0.2%. These differences are consistent with the corresponding variations in the extracted FSR and therefore do not indicate a systematic offset between the two measurement setups. The remaining differences are attributed to slight fitting and measurement variations.

TABLE IV
COMPARISON OF GROUP INDICES EXTRACTED FROM IRIS AND SCYLLA MEASUREMENTS AT 1550 NM. THE RELATIVE DEVIATION IS DEFINED WITH RESPECT TO THE IRIS DATASET.

Device	ΔL (μm)	$n_{g,\text{Iris}}$	$n_{g,\text{Scylla}}$	Deviation (%)
MZI1	30	4.2091	4.2008	-0.20
MZI2	60	4.1785	4.1832	+0.11
MZI3	90	4.1869	4.1876	+0.02
MZI4	90	4.1885	4.1915	+0.07
MZI5	90	4.1912	4.1917	+0.01
MZI6	120	4.1895	4.1921	+0.06
MZI7	150	4.1878	4.1891	+0.03
MZI8	190	4.2313	4.2322	+0.02

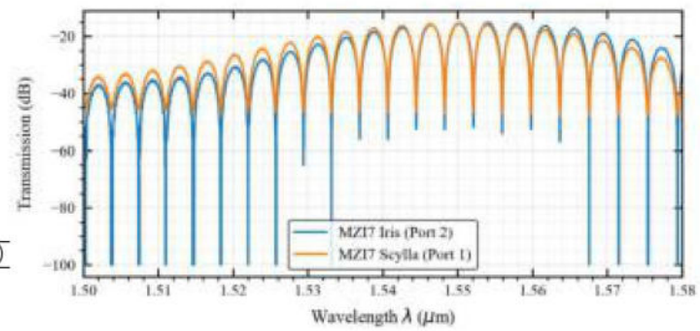

Fig. 17. Comparison of the raw data from the Iris stage compared to the Scylla stage for MZI7. The Iris measurements show broader fringes at shorter wavelengths and a larger variety of fringe depth compared to the Scylla stage.

The results from both measurement sets are within the range calculated during the corner analysis and show similar relative device characteristics as shown in figure 18

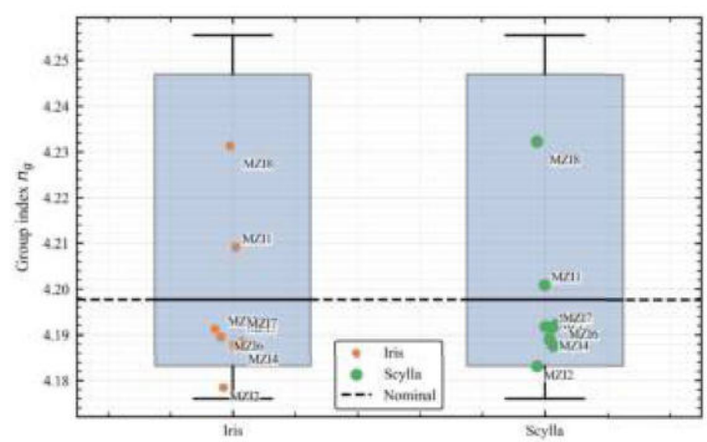

Fig. 18. Box-plot comparison of the extracted group indices of the characterized devices for each measurement data set compared to the values simulated using corner analysis. All extracted group indices are within the range of the corner analysis.

TABLE V
COMPARISON OF EXTRACTED FSR AT 1550 NM FOR THE IRIS AND SCYLLA DATA SETS.

Device	ΔL (μm)	FSR Iris (nm)	FSR Scylla (nm)	Rel. (%)
MZI1	30	19.01	19.05	+0.20
MZI2	60	9.58	9.57	-0.11
MZI3	90	6.37	6.37	-0.02
MZI4	90	6.37	6.36	-0.07
MZI5	90	6.37	6.37	-0.01
MZI6	120	4.78	4.78	-0.06
MZI7	150	3.82	3.82	-0.03
MZI8	190	3.01	3.01	-0.02

IX. CONCLUSION

Unbalanced Mach–Zehnder interferometers with path length differences between 30 µm and 190 µm were designed, simulated, fabricated, and experimentally characterized to extract the group index of a quasi-TE mode in a silicon strip waveguide. The nominal group index obtained from simulation at 1550 nm is $n_g = 4.1976$.

The experimentally extracted values (Scylla) range from $n_g = 4.1832$ to 4.2322, corresponding to deviations from the nominal simulation between -0.35% and +0.82%. For most devices, the deviation remains below 0.25% and is negative. A reduced group index suggests slightly increased waveguide dimensions, consistent with reduced dispersion. All measured values lie within the range predicted by corner analysis (n_g ≈4.18–4.26), demonstrating good agreement between simulation and experiment and confirming that the observed variations are consistent with fabrication tolerances.

The three devices with the same path length difference of $\Delta L = 90\,\mu m$ exhibit a spread of 0.0041 in n_g ($\sim 0.10\%$), indicating good reproducibility. A comparison between the Iris and Scylla datasets shows relative deviations below 0.2%, consistent with the corresponding variations in the extracted FSR. All values of the group index are within the theoretical bounds calculated from the corner analysis. Overall, the results confirm that unbalanced MZIs provide a robust method for group index extraction.

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